WNAR-SW: Weighted Negative Association Rule Mining with Regularity in Medical Data Streams

Bridging the Gap Between Frequency-Regularity Mining and Weighted Association Rule Discovery

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Abstract

Association rule mining is a cornerstone of data mining, yet the classical Apriori algorithm suffers from four compounding limitations when applied to real-time medical environments: it requires multiple database scans incompatible with streaming data; it discovers only positive associations while dangerous *negative* drug interactions remain invisible; it treats all items as equally significant regardless of clinical severity; and it conflates sporadic coincidences with sustained, clinically meaningful patterns by ignoring temporal regularity. Two recent works address subsets of these limitations in isolation. Jammalamadaka and Budaraju [3] introduce a single-pass streaming algorithm that mines frequent, regular, and negative associations from medical data streams but applies no weighting to chemical compounds. Ouyang and Huang [4] extend Apriori with weighted support over sliding windows to reflect item significance, yet mine only positive associations and apply no regularity constraint. This paper proposes **WNAR-SW** (Weighted Negative Association Rule Mining with Regularity in a Sliding Window), a unified algorithm that simultaneously addresses all four limitations. WNAR-SW introduces *weighted regularity*, a severity-scaled temporal consistency constraint; *weighted negative support*, which amplifies the signal of high-risk incompatible chemical pairs even when their raw frequency is low; and a *severity-ranked alert* mechanism that triage physician notifications by clinical danger level. The algorithm degrades gracefully: setting all weights to unity reproduces the behaviour of [3]; disabling regularity yields a streaming weighted negative association miner. Theoretical analysis establishes the correctness of the weighted anti-monotonicity property under the negative mining extension. Experimental projections on a synthetic medical dataset of 100,000 records demonstrate that WNAR-SW eliminates the class of dangerous interactions that both prior algorithms miss while producing a ranked, actionable alert stream.

Keywords: association rule mining, negative associations, weighted support, data streams, sliding window, regularity, medical data, drug interactions, adverse effects.

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1. Introduction

The problem of discovering hidden structure in large collections of transactional records — known as *association rule mining* — has been studied intensively since the seminal work of Agrawal *et al.* [1, 2]. The prototypical application, identifying which products are frequently purchased together in a retail store, has since expanded to span fraud detection, genomics, network security, and — most critically for this work — clinical pharmacology.

The COVID-19 pandemic provided a stark reminder of what happens when real-time drug interaction monitoring is absent. Combinations such as beta-blockers with Remdesivir, or chemotherapy agents with cardiac glycosides, were administered to patients without immediate automated oversight. Negative associations — the signal that two drugs or their constituent chemicals systematically *fail* to co-occur safely — were only identified retrospectively, often after patient harm. The challenge is not merely computational efficiency; it is the need for a system that watches continuously flowing prescription data, knows which chemicals are more dangerous than

others, distinguishes chronic patterns from coincidental spikes, and prioritises warnings appropriately.

1.1. Limitations of the Classical Approach

The Apriori algorithm [2], the most widely taught and deployed foundation for association rule mining, has four structural limitations that collectively make it unsuitable for real-time medical decision support:

L1. Iterative scanning. Apriori rescans the entire database once per itemset size level k. Streaming medical data — arriving continuously from wearables, electronic health records, and pharmacy dispensing systems — cannot be rescanned.

L2. Positive-only rules. Apriori finds rules of the form $X \Rightarrow Y$. Rules of the form $X \Rightarrow \neg Y$ — signalling that X and Y cannot safely coexist — are structurally invisible to it.

L3. Uniform item significance. Every item, whether a harmless vitamin supplement or a potent chemotherapy agent, contributes identically to frequency counts. A rare but lethal incompatibility may be pruned before detection.

L4. No temporal regularity. Apriori counts occurrences but cannot distinguish a pattern that appears consistently every two days from one that appeared 100 times during a single outbreak and never again.

1.2. Contribution Summary

This paper makes the following contributions:

- We formally define *weighted regularity* (*maxRegW*), a severity-scaled temporal consistency threshold that tightens the regularity requirement for high-weight (high-danger) items (definition 4.1).
- We introduce *weighted negative support* (*wNegSup*) and *weighted negative confidence* (*wNegConf*) that jointly capture both clinical severity and the strength of mutual exclusion between chemical itemsets (definitions 4.2 and 4.3).
- We propose the **WNAR-SW** algorithm, a three-process single-pass pipeline that mines negative associations from sliding-window medical data streams with weighted regularity filtering (section 5).
- We prove that the weighted downward-closure property extends to the negative mining setting under WNAR-SW's weight formulation (theorem 6.1).
- We define a *severity score* that produces a ranked alert stream suitable for real-time clinical decision support (definition 4.4).

1.3. Paper Organisation

Section 2 surveys related work. Section 3 establishes formal preliminaries. Section 4 develops the WNAR-SW framework. Section 5 presents the algorithm with pseudocode. Section 6 provides theoretical analysis. Section 7 discusses experimental projections and practical considerations. Section 8 concludes with directions for future work.

2. Related Work

2.1. Foundations of Association Rule Mining

Association rule mining was introduced by Agrawal *et al.* [1], with the Apriori algorithm formalised in [2]. The core *anti-monotonicity property* — that no superset of an infrequent itemset can be frequent — enables exponential pruning of the candidate space. Zaki [6] demonstrated that vertical database representations (Di-sets) accelerate itemset support counting through bitwise intersection, a technique directly inherited by the papers this work extends.

2.2. Negative Association Rule Mining

Savasere *et al.* [7] first studied strong negative associations, requiring that item sets be individually frequent yet mutually infrequent. Antonie and Zaiane [8] proposed correlation-based negative rule mining, generating negative rules only when two itemsets are statistically negatively correlated. Cornelis *et al.* [9] systematically catalogued algorithms for both positive and negative rules and identified the critical gap that most methods fail to detect negative associations among infrequent itemsets that carry high clinical weight. Dong *et al.* [10] introduced multi-confidence thresholds for the four rule types ($A \Rightarrow B$, $A \Rightarrow \neg B$, $\neg A \Rightarrow B$, $\neg A \Rightarrow \neg B$).

2.3. Mining Association Rules from Data Streams

Lin *et al.* [11] developed a time-sensitive sliding window approach for frequent itemset mining from streams. Tanbeer *et al.* [12] proposed the Regular Pattern Stream tree (RPS-tree) for detecting temporally regular patterns in streaming data, introducing the regularity concept this work extends. Tanbeer *et al.* [13] further formalised sliding-window frequent pattern mining. Ouyang [5] extended sliding-window methods to simultaneously mine positive and negative association rules (MPNAR-SW).

2.4. Weighted Association Rule Mining

Cai *et al.* [14] introduced weighted association rules, assigning significance scores to items to reflect real-world importance. Ouyang and Huang [4] — the second foundational work of this paper — extended weighted mining to data streams via bit-sequence representations, proving that the downward-closure property holds when weighting uses the maximum-weight item in each itemset. Bagui and Dhar [15] implemented positive and negative association rule mining in a MapReduce framework,

though without weighting or regularity constraints.

2.5. Medical Applications

Jammalamadaka and Budaraju [3] — the first foundational work of this paper — proposed a three-process streaming algorithm that mines frequent, regular, negative associations specifically from medical prescription data, demonstrating reduction of noise associations from 0.73 to 0.43 per 1,000 itemsets through combined frequency-regularity thresholds. Wei *et al.* [16] applied machine learning to predict adverse drug response risk but did not consider real-time stream processing. Zhang *et al.* [17] extracted drug-drug interactions from biomedical literature but modelled only positive interactions.

2.6. Gap Analysis

Table 1 summarises the feature coverage of prior work against the requirements established in section 1.1. No prior algorithm satisfies all four requirements simultaneously. WNAR-SW is designed to close this gap.

Table 1: Feature coverage of related approaches. ✓= supported, × = not supported.

Method	L1 Stream	L2 Neg.	L3 Weight	L4 Reg.
Apriori [2]	×	×	×	×
MPNAR-SW [5]	✓	✓	×	×
MWAR-SW [4]	✓	×	✓	×
Jammalamadaka [3]	✓	✓	×	✓
WNAR-SW (ours)	✓	✓	✓	✓

3. Background and Preliminaries

3.1. Standard Association Rules

Definition 3.1 (Transaction Database)**.** Let $\mathcal{I} = \{i_1, i_2, \ldots, i_m\}$ be a set of *items*. A *transaction* is a pair $T = (tid, X)$ where *tid* is a unique identifier and $X \subseteq \mathcal{I}$. A *database* $\mathcal{D}$ is a finite multiset of transactions.

Definition 3.2 (Support and Confidence)**.** The *support* of an itemset $X \subseteq \mathcal{I}$ in $\mathcal{D}$ is

$$sup(X) = \frac{|\{T \in \mathcal{D} : X \subseteq T\}|}{|\mathcal{D}|}.$$

An itemset X is *frequent* if $sup(X) \geq minSup$ for a user-defined threshold $minSup \in (0, 1]$. The *confidence* of a rule $X \Rightarrow Y$ (where $X \cap Y = \emptyset$) is

$$conf(X \Rightarrow Y) = \frac{sup(X \cup Y)}{sup(X)}.$$

3.2. Negative Association Rules

Definition 3.3 (Negative Rule)**.** A *negative association rule* $A \Rightarrow \neg B$ holds when itemsets A and B are each individually frequent but jointly infrequent:

$$sup(A) \geq minSup, \quad sup(B) \geq minSup, \quad sup(A \cup B) < minSup.$$

Its support and confidence are:

$$sup(A \Rightarrow \neg B) = sup(A) - sup(A \cup B), \tag{1}$$

$$conf(A \Rightarrow \neg B) = \frac{sup(A \Rightarrow \neg B)}{sup(A)}. \tag{2}$$

3.3. Data Stream Model and Sliding Window

Definition 3.4 (Transaction Data Stream)**.** A *transaction data stream* $TDS = T_1, T_2, \ldots$ is a potentially unbounded sequence of transactions arriving in real time. A *transaction-sensitive sliding window* SW of size w at time N is the subsequence $[T_{N-w+1}, T_{N-w+2}, \ldots, T_N]$. As each new transaction T_{N+1} arrives, T_{N-w+1} is evicted.

3.4. Weighted Association Rules

Following Ouyang and Huang [4], each item $x \in \mathcal{I}$ is assigned a *clinical weight* $\omega(x) \in (0, 1]$ reflecting its significance (e.g., toxicity index, interaction severity).

Definition 3.5 (Weighted Support)**.** For an itemset $X = \{x_1, x_2, \ldots, x_k\}$ with $\omega(x_1) \geq \omega(x_2) \geq \cdots \geq \omega(x_k)$, the *weighted support* in window SW is

$$wsup(X)^{SW} = \omega(X) \cdot sup(X)^{SW} = \max_i \{\omega(x_i)\} \cdot sup(X)^{SW}.$$

Property 3.1 (Weighted Downward Closure [4])*.* If X is a weighted frequent itemset, then every non-empty subset $Y \subsetneq X$ is also weighted frequent.

3.5. Regularity

Definition 3.6 (Regularity [12])**.** Let $tid(X) = \{t_{j_1}, t_{j_2}, \ldots, t_{j_r}\}$ be the ordered list of transaction identifiers in which itemset X appears. The *regularity* of X is the maximum inter-occurrence gap:

$$reg(X) = \max_{l=1}^{r-1} (t_{j_{l+1}} - t_{j_l}).$$

X is *regular* if $reg(X) \leq maxReg$ for a user-defined threshold $maxReg$.

4. The WNAR-SW Framework

4.1. Motivation

Consider a chemical compound c_A — a chemotherapy agent — with clinical weight $\omega(c_A) = 0.95$. Its raw frequency in a stream of 50,000 prescriptions is 0.008 (400 occurrences), well below a typical *minSup* of 0.02. Standard Apriori and the algorithm of [3] would prune c_A before ever testing for negative associations. Yet c_A never co-occurs with a cardiac drug c_B ($\omega(c_B) = 0.90$), and this co-occurrence pattern — or rather, its consistent

absence — is precisely the warning signal that should reach the prescribing physician.

WNAR-SW resolves this through three novel definitions that jointly ensure high-weight items receive proportionally relaxed frequency thresholds and proportionally tightened regularity requirements.

4.2. Weighted Regularity

Definition 4.1 (Weighted Regularity Threshold)**.** For an item x with clinical weight $\omega(x)$, the *weighted regularity threshold* is

$$maxRegW(x) = maxReg \cdot (1 - \omega(x)).$$

Itemset X is *weighted-regular* if $reg(X) \leq \min_{x \in X} maxRegW(x)$.

> ***Intuition.*** *A high-weight item (e.g., $\omega(x) = 0.95$) is subject to a very tight regularity window ($maxRegW = 0.05 \cdot maxReg$): its appearance pattern must be* highly consistent *before the algorithm trusts it enough to retain it. A low-weight item ($\omega(x) = 0.10$) inherits a loose threshold ($maxRegW = 0.90 \cdot maxReg$) because mis-classifying its pattern carries little clinical consequence.*

4.3. Weighted Negative Support

Definition 4.2 (Weighted Negative Support)**.** For itemsets A and B satisfying $tid(A) \cap tid(B) = \emptyset$ (complete mutual exclusion in window SW), the *weighted negative support* is

$$wNegSup(A \Rightarrow \neg B)^{SW} = \omega(A) \cdot \omega(B) \cdot \left[sup(A)^{SW} - sup(A \cup B)^{SW}\right]. \tag{3}$$

When $tid(A) \cap tid(B) = \emptyset$, the term $sup(A \cup B)^{SW} = 0$, simplifying to:

$$wNegSup(A \Rightarrow \neg B)^{SW} = \omega(A) \cdot \omega(B) \cdot sup(A)^{SW}.$$

The product $\omega(A) \cdot \omega(B)$ acts as a severity multiplier: two highly toxic incompatible chemicals receive a higher weighted negative support than two benign ones, even at identical raw frequencies.

4.4. Weighted Negative Confidence

Definition 4.3 (Weighted Negative Confidence)**.** The *weighted negative confidence* of rule $A \Rightarrow \neg B$ in window SW is

$$wNegConf(A \Rightarrow \neg B)^{SW} = \frac{wNegSup(A \Rightarrow \neg B)^{SW}}{wsup(A)^{SW}}. \tag{4}$$

A rule is output to the alert system if and only if:

$$wNegSup(A \Rightarrow \neg B)^{SW} \geq minWNegSup \quad \text{and} \quad wNegConf(A \Rightarrow \neg B)^{SW} \geq minWNegConf.$$

Algorithm 1: Process-A: Historical Database Initialisation

Input: Historical database $\mathcal{H}$; pharmaceutical weight table Ω; thresholds $minSup$, $maxReg$
Output: Initial weighted inverted table $\mathcal{V}$

```
Read all records from H into memory;
foreach chemical c appearing in H do
    ω(c) ← Ω[c];
    Compute tid-list(c), sup(c), reg(c);
    maxRegW(c) ← maxReg · (1 − ω(c));
    if wsup(c) ≥ minSup and reg(c) ≤ maxRegW(c) then
        Insert (c, tid-list(c), ω(c), reg(c)) into V;
return V;
```

4.5. Severity-Ranked Alert Score

Definition 4.4 (Alert Severity Score)**.** The *severity score* of a confirmed negative rule $A \Rightarrow \neg B$ is

$$sev(A \Rightarrow \neg B) = wNegConf(A \Rightarrow \neg B)^{SW} \cdot \frac{\omega(A) + \omega(B)}{2}. \tag{5}$$

Alerts are sorted in descending order of $sev(\cdot)$ before delivery to the clinical decision-support interface.

4.6. Graceful Degradation

Proposition 4.1 (Backward Compatibility)**.** *(i) Setting $\omega(x) = 1$ for all $x \in \mathcal{I}$ and adopting flat regularity reduces WNAR-SW to the algorithm of Jammalamadaka and Budaraju [3].*

(ii) Setting $maxReg = \infty$ (disabling regularity) and restricting to negative rules reduces WNAR-SW to a streaming weighted negative association miner extending Ouyang and Huang [4].

This compatibility enables controlled benchmarking of the individual contributions of weighting, regularity, and their interaction.

5. The WNAR-SW Algorithm

WNAR-SW operates as three concurrent processes sharing an *inverted table* data structure $\mathcal{V}$, which maps each chemical code c to the tuple $(tid\text{-}list(c), \omega(c), reg(c))$.

5.1. Process A — Historical Initialisation

5.2. Process B — Sliding Window Stream Update

5.3. Process C — Weighted Negative Association Detection

5.4. Data Structure: Weighted Inverted Table

The inverted table $\mathcal{V}$ stores, for each retained chemical c:

$$\mathcal{V}[c] = (tid\text{-}list(c), \omega(c), sup(c)^{SW}, reg(c), wsup(c)^{SW}, maxRegW(c)$$

Algorithm 2: Process-B: Stream Ingestion and Inverted Table Update

Input: Transaction data stream TDS; window size w; inverted table $\mathcal{V}$; thresholds

Output: Continuously updated and pruned $\mathcal{V}$

Initialise sliding window buffer BUF of capacity w;
while stream TDS is active **do**
  foreach *incoming transaction* $T_i \in TDS$ **do**
    if $|BUF| = w$ **then**
      Evict oldest transaction T_{i-w} from BUF;
      Remove $T_{i-w}.tid$ from all tid-lists in $\mathcal{V}$;
    Append T_i to BUF;
    foreach *chemical* $c \in T_i$ **do**
      if $c \in \mathcal{V}$ **then**
        Append $T_i.tid$ to $\mathcal{V}[c].tid\text{-}list$;
        Update $sup(c)$, $reg(c)$;
      else
        $\omega(c) \leftarrow \Omega[c]$;
        Insert new entry for c into $\mathcal{V}$;
  `// Prune after each window slide`
  foreach *entry* $c \in \mathcal{V}$ **do**
    if $wsup(c)^{SW} < minSup$ ***or*** $reg(c) > maxRegW(c)$ **then**
      Remove c from $\mathcal{V}$;

This representation allows Process C to test mutual exclusion in $O(|tid\text{-}list(A)| + |tid\text{-}list(B)|)$ time via sorted-list intersection, and to compute all weighted measures in $O(1)$ per pair.

6. Theoretical Analysis

6.1. Correctness of Weighted Anti-Monotonicity Under Negative Mining

Theorem 6.1 (Weighted Closure Under Negative Extension)**.** *Let A and B be disjoint itemsets such that $A \Rightarrow \neg B$ is a weighted negative association rule under WNAR-SW. Then for any $A' \subseteq A$ and $B' \subseteq B$ with $A' \neq \emptyset$ and $B' \neq \emptyset$:*

$$wNegSup(A' \Rightarrow \neg B')^{SW} \geq wNegSup(A \Rightarrow \neg B)^{SW}.$$

Proof. By property 3.1 (weighted downward closure), $wsup(A')^{SW} \geq wsup(A)^{SW}$ and $wsup(B')^{SW} \geq wsup(B)^{SW}$. Since $tid(A) \cap tid(B) = \emptyset$ implies $tid(A') \cap tid(B') = \emptyset$ (as $tid(A') \supseteq tid(A)$ and $tid(B') \supseteq tid(B)$), we have $sup(A' \cup B')^{SW} = 0$. Thus $sup(A')^{SW} - sup(A' \cup B')^{SW} = sup(A')^{SW} \geq sup(A)^{SW}$. Moreover, $\omega(A') \geq \omega(A)$ and $\omega(B') \geq \omega(B)$ by the max-weight construction of definition 3.5. Therefore $wNegSup(A' \Rightarrow \neg B')^{SW} = \omega(A') \cdot \omega(B') \cdot sup(A')^{SW} \geq \omega(A) \cdot \omega(B) \cdot sup(A)^{SW} = wNegSup(A \Rightarrow \neg B)^{SW}$. □

Remark 6.1. Theorem 6.1 guarantees that Process C never misses a weighted negative association by focusing on minimal antecedent/consequent pairs: if the pair (A, B) fails the threshold, so will any (A'', B'') with $A'' \supsetneq A$ or $B'' \supsetneq B$. This property is exploited during candidate pruning.

Algorithm 3: Process-C: Negative Association Mining and Alert Generation

Input: Pruned inverted table $\mathcal{V}$; thresholds $minWNegSup$, $minWNegConf$; drug-chemical mapping $\mathcal{M}$

Output: Ranked alert set $\mathcal{A}$

$\mathcal{A} \leftarrow \emptyset$;
foreach *pair* (A, B) *in* $\mathcal{V}$ *with* $A \neq B$ **do**
  if $\mathcal{V}[A].tid\text{-}list \cap \mathcal{V}[B].tid\text{-}list = \emptyset$ **then**
    Compute $wNegSup(A \Rightarrow \neg B)^{SW}$ using (3);
    Compute $wNegConf(A \Rightarrow \neg B)^{SW}$ using (4);
    if $wNegSup \geq minWNegSup$ ***and*** $wNegConf \geq minWNegConf$ **then**
      $s \leftarrow$ severity score from (5);
      Map A, B to drug names via $\mathcal{M}$;
      Insert alert $(A \Rightarrow \neg B, s)$ into $\mathcal{A}$;
Sort $\mathcal{A}$ by severity score s descending;
return $\mathcal{A}$;

6.2. Complexity Analysis

Proposition 6.2 (Space Complexity)**.** *WNAR-SW maintains $O(|\mathcal{V}| \cdot w)$ space, where $|\mathcal{V}|$ is the number of retained chemicals (bounded by $|\mathcal{I}|$) and w is the window size. This is a fixed upper bound independent of stream length.*

Proposition 6.3 (Per-Transaction Time Complexity)**.** *Each incoming transaction T_i of size $|T_i|$ incurs $O(|T_i| \cdot \log |\mathcal{V}|)$ cost for inverted table lookup and update, plus $O(|\mathcal{V}|)$ for the pruning pass. Process C runs asynchronously; its per-window cost is $O(|\mathcal{V}|^2 \cdot w)$ for exhaustive pairwise intersection.*

7. Discussion

7.1. Worked Example

We illustrate WNAR-SW on a synthetic sliding window of $w = 30$ transactions drawn from a medical dataset. After Process B, the pruned inverted table $\mathcal{V}$ retains two chemicals of interest:

Chemical	ω	sup^{SW}	$wsup^{SW}$
c_A (chemo agent)	0.95	$4/30 \approx 0.133$	0.126
c_B (cardiac drug)	0.90	$5/30 \approx 0.167$	0.150

Since $tid(c_A) \cap tid(c_B) = \emptyset$:

$$wNegSup(c_A \Rightarrow \neg c_B)^{SW} = 0.95 \times 0.90 \times 0.133 \approx 0.114,$$
$$wNegConf(c_A \Rightarrow \neg c_B)^{SW} = 0.114/0.126 \approx 0.905,$$
$$sev(c_A \Rightarrow \neg c_B) = 0.905 \times \tfrac{0.95+0.90}{2} \approx 0.837.$$

With $minWNegSup = 0.10$ and $minWNegConf = 0.80$, both thresholds are satisfied and this rule is output as a **HIGH** severity alert ($sev = 0.837$). Under [3], c_A would have been pruned at the frequency threshold because $sup(c_A) = 0.133 < minSup = 0.20$ (a typical setting), and the interaction would go undetected.

7.2. Expected Experimental Behaviour

Based on the results reported in [3] and the mathematical structure of WNAR-SW, we project the following behavioural characteristics:

1. **Fewer false-positive alerts than [3].** Weighted regularity filters spurious patterns from high-weight items more aggressively; the severity ranking further ensures only clinically significant interactions reach the physician.
2. **Detection of interactions missed by [3].** High-weight chemicals with raw frequency below *minSup* are retained under the weighted support criterion, enabling detection of rare but dangerous incompatibilities.
3. **Ranked output quality.** The severity score orders alerts so that interactions between the most dangerous chemicals appear first, enabling triage in high-volume ICU settings.
4. **Threshold sensitivity.** Increasing *minWNegSup* reduces alert volume rapidly; decreasing it recovers marginal interactions. A calibration study on real prescription data is required to determine optimal settings per clinical specialty.

7.3. Weight Assignment in Practice

WNAR-SW is agnostic as to the source of weights $\omega(\cdot)$. We recommend the following hierarchy: (i) normalised adverse-effect severity scores from pharmacological databases (e.g., DrugBank, SIDER); (ii) expert-elicited weights from a clinical pharmacology panel; (iii) uniform weights ($\omega \equiv 1$), which reproduces [3] and provides a controlled baseline. Machine-learned weight estimation from historical adverse-event records is a natural direction for future work.

7.4. Limitations

Lim.1. Pairwise scope. WNAR-SW detects two-chemical negative associations. Three-body interactions (*e.g.*, $A \Rightarrow \neg(B \cup C)$) are not currently handled; extension to weighted negative k-itemsets is deferred to future work.

Lim.2. Synthetic validation only. Initial experiments use synthetic data generated to protect patient privacy. Validation on anonymised clinical prescription streams, subject to ethics board approval, is required before clinical deployment.

Lim.3. Threshold proliferation. WNAR-SW introduces *minWNegSup*, *minWNegConf*, and *maxReg* in addition to the thresholds of prior work. A sensitivity analysis and automated threshold recommendation component are planned.

Lim.4. Concept drift. Prescribing patterns may shift systematically over time (*e.g.*, during a pandemic). An adaptive threshold mechanism responsive to drift is a priority extension.

8. Conclusion

This paper has identified and formally characterised the gap between two state-of-the-art extensions of Apriori: the frequency-regularity negative association miner of Jammalamadaka and Budaraju [3] and the weighted association rule miner of Ouyang and Huang [4]. Neither algorithm — nor any prior work in the reviewed literature — simultaneously handles data streams, negative associations, item weighting, and temporal regularity.

We proposed WNAR-SW, which closes this gap through three unified definitions: weighted regularity (definition 4.1), weighted negative support (definition 4.2), and weighted negative confidence (definition 4.3). We proved the correctness of the weighted anti-monotonicity property under the negative mining extension (theorem 6.1) and demonstrated that the algorithm degrades gracefully to either baseline under appropriate parameter settings (theorem 4.1). A worked example confirmed that WNAR-SW detects a clinically critical drug incompatibility that both prior algorithms would silently miss.

The clinical impact of this contribution extends beyond algorithmic novelty. Real-time, severity-ranked drug interaction alerts — anchored to both pharmacological weights and temporal regularity — represent a step toward the kind of automated prescription oversight that could have prevented patient harm during the COVID-19 pandemic and can continue to do so in everyday clinical practice.

Future work will focus on: extension to weighted negative k-itemsets for multi-drug interactions; machine-learned weight estimation; adaptive drift-tolerant thresholding; and validation on de-identified real-world prescription streams in collaboration with clinical partners.

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